

Perfect. After the DDD, the next document is:

Backend API Specification Document (BASD)

ScrapFlow AI – Version 1.0

1. Document Overview

Purpose

This document defines:

- REST API endpoints
- Request structures
- Response structures
- Validation rules
- Authentication requirements
- Error handling

This document will be used by:

- Backend Developers
 - Frontend Developers
 - API Testing
 - Future Mobile Applications
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2. API Architecture

Frontend



REST API



FastAPI



MySQL

3. Authentication APIs

Base URL

/api/v1/auth

Register User

Endpoint

POST /api/v1/auth/register

Request

```
{  
  "name": "Zaid",  
  "email": "zaid@example.com",  
  "password": "password123"  
}
```

Response

```
{  
  "message": "User registered successfully",  
  "user_id": 1  
}
```

Login

Endpoint

POST /api/v1/auth/login

Request

```
{
```

```
"email": "zaid@example.com",  
"password": "password123"  
}
```

Response

```
{  
  "access_token": "jwt_token",  
  "token_type": "bearer"  
}
```

Get Current User

Endpoint

GET /api/v1/auth/me

Headers

Authorization: Bearer <token>

Response

```
{  
  "id": 1,  
  "name": "Zaid",  
  "email": "zaid@example.com"  
}
```

4. Batch APIs

Base URL

/api/v1/batches

Create Batch

Endpoint

POST /api/v1/batches

Request

```
{  
  "batch_name": "BATCH-001",  
  "collection_date": "2026-06-04",  
  "collection_location": "Nagpur",  
  "collection_cost": 5000,  
  "fuel_cost": 500  
}
```

Response

```
{  
  "message": "Batch created",  
  "batch_id": 1  
}
```

Get All Batches

Endpoint

GET /api/v1/batches

Get Batch By ID

Endpoint

GET /api/v1/batches/{id}

Update Batch

Endpoint

PUT /api/v1/batches/{id}

Delete Batch

Endpoint

DELETE /api/v1/batches/{id}

5. Materials APIs

Base URL

/api/v1/materials

Add Material

Endpoint

POST /api/v1/materials

Request

```
{  
  "batch_id": 1,  
  "material_type": "Copper",  
  "weight": 20  
}
```

Response

```
{  
  "message": "Material added"  
}
```

Get Materials

Endpoint

GET /api/v1/materials

Update Material

Endpoint

PUT /api/v1/materials/{id}

Delete Material

Endpoint

DELETE /api/v1/materials/{id}

6. Inventory APIs

Base URL

/api/v1/inventory

Get Inventory

Endpoint

GET /api/v1/inventory

Response

```
[  
  {  
    "material_type": "Copper",  
    "available_weight": 120  
  }  
]
```

Get Inventory Summary

Endpoint

GET /api/v1/inventory/summary

Response

```
{  
  "total_weight": 500,  
  "inventory_value": 250000  
}
```

7. Buyer APIs

Base URL

/api/v1/buyers

Create Buyer

POST /api/v1/buyers

Request

```
{  
  "name": "ABC Metals",  
  "phone": "9876543210",  
  "address": "Nagpur"  
}
```

Get Buyers

GET /api/v1/buyers

Update Buyer

PUT /api/v1/buyers/{id}

Delete Buyer

DELETE /api/v1/buyers/{id}

8. Sales APIs

Base URL

/api/v1/sales

Create Sale

Endpoint

POST /api/v1/sales

Request

```
{  
  "buyer_id": 1,  
  "material_id": 2,  
  "quantity": 50,  
  "price_per_kg": 750  
}
```

Response

```
{  
  "message": "Sale created",  
  "total_amount": 37500  
}
```

Get Sales

GET /api/v1/sales

Get Sale By ID

GET /api/v1/sales/{id}

9. Expense APIs

Base URL

/api/v1/expenses

Create Expense

POST /api/v1/expenses

Request

```
{  
  "batch_id": 1,  
  "expense_type": "Fuel",  
  "amount": 500  
}
```

Get Expenses

GET /api/v1/expenses

Delete Expense

DELETE /api/v1/expenses/{id}

10. Dashboard APIs

Base URL

/api/v1/dashboard

Dashboard Summary

Endpoint

GET /api/v1/dashboard/summary

Response

```
{  
  "today_sales": 15000,  
  "today_expenses": 5000,  
  "today_profit": 10000,  
  "inventory_value": 250000  
}
```

11. Reports APIs

Base URL

/api/v1/reports

Profit Report

GET /api/v1/reports/profit

Inventory Report

GET /api/v1/reports/inventory

Sales Report

GET /api/v1/reports/sales

Export PDF

GET /api/v1/reports/pdf

Export Excel

GET /api/v1/reports/excel

12. Validation Rules

User

Email must be unique.

Password minimum 8 characters.

Batch

Collection cost ≥ 0

Fuel cost ≥ 0

Material

Weight > 0

Sale

Quantity > 0

Price > 0

Inventory must exist

Expense

Amount > 0

13. Error Responses

400

```
{  
  "error": "Invalid request data"  
}
```

401

```
{  
  "error": "Unauthorized"  
}
```

404

```
{  
  "error": "Resource not found"  
}
```

500

```
{  
  "error": "Internal server error"  
}
```

14. Authentication Flow

Register



Login



JWT Generated



Frontend Stores Token



Protected APIs Accessible

15. Inventory Flow

Create Batch



Add Material



Inventory Updated

16. Sales Flow

Create Sale



Check Inventory



Reduce Inventory



Store Sale



Update Dashboard

17. API Development Priority

Phase 1

Auth APIs

Batch APIs

Material APIs

Phase 2

Inventory APIs

Buyer APIs

Sales APIs

Phase 3

Dashboard APIs

Report APIs
